Joint Statement of Joint Stakeholder Proposal On Recommended Energy Conservation Standards And Test Procedure For Consumer Room Air Cleaners

August 23, 2022

Association of Home Appliance Manufacturers[1]
Appliance Standard Awareness Project
American Council for an Energy-Efficient Economy
Consumer Federation of America
Natural Resources Defense Council
New York State Energy Research and Development Authority
Pacific Gas and Electric Company

I. <u>Introduction And Overview</u>

The Association of Home Appliance Manufacturers (AHAM),[1] Appliance Standards Awareness Project (ASAP), American Council for an Energy-Efficient Economy (ACEEE), Consumer Federation of America (CFA), Natural Resources Defense Council (NRDC), the New York State Energy Research and Development Authority (NYSERDA), and the Pacific Gas and Electric Company (PG&E) (collectively, Joint Stakeholders) are pleased to present to the U.S. Department of Energy (the Department or DOE) the results of successful negotiations which resulted in an agreement (the Joint Proposal) on federal minimum energy conservation standards for consumer room air cleaners and the related test procedures. This agreement was aided by DOE's contractor's engagement with the AHAM Task Force responsible for the air cleaner test procedure.

Central to this agreement are the proposed timelines for compliance. The Joint Stakeholders urge DOE to publish final rules adopting the consumer room air cleaner test procedure and standards and compliance dates contained in the Joint Proposal, as soon as possible, but not later than December 31, 2022. If DOE does not meet that deadline, each of the Joint Stakeholders reserve the right to rescind support for the standards and compliance dates in the Joint Proposal.

The Joint Proposal is fully consistent with the requirements of the Energy Policy and Conservation Act of 1975, as amended (EPCA) with regard to energy conservation standards and applicable test procedures. And it satisfies the criteria for DOE to use its direct final rule authority in 42 U.S.C. § 6295(p)(4) to issue a direct final rule to establish energy conservation standards. As discussed more fully below:

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[1] Representing the following companies who manufacture consumer room air cleaners and are members of the Portable Appliance Division: 3M Co., ACCO Brands Corporation, Airgle Corporation, Alticor, Inc., Beijing Smartmi Electronic Technology Co., Ltd., BISSELL Inc., Blueair Inc., BSH Home Appliances Corporation, De'Longhi America, Inc., Dyson Limited, Essick Air Products, Fellowes Inc., Foxconn Technology Group, Gree Electric Appliances Inc., Groupe SEB, Haier Smart Home Co., Ltd., Helen of Troy-Health & Home, Lasko Products, Inc., Molekule Inc., Newell Brands Inc., Oransi LLC, Phillips Domestic Appliances NA Corporation, SharkNinja Operating, LLC, Vornado Air LLC, Winix Inc., and Zojirushi America Corporation.

- The Joint Stakeholders are representative of a wide range of expert and relevant points of view—including those of manufacturers of various sizes and representing over sixty percent of the market; States; and consumer, environmental, and efficiency advocacy groups.
- The recommended test procedure is designed to meet statutory requirements in 42 U.S.C. § 6293(b)(3).
- The recommended energy conservation standards are designed to achieve the maximum improvement in energy efficiency that is technologically feasible and economically justified in accordance with the provisions of 42 U.S.C. § 6295(o).

Accordingly, we strongly urge the Department to finalize rules that adopt the recommendations in the Joint Proposal as soon as possible and no later than December 31, 2022.

II. The Joint Stakeholders Are Fairly Representative Of Relevant Points Of View.

EPCA permits DOE to adopt new standards via a direct final rule on receipt of a statement that is submitted jointly by interested persons that are fairly representative of the relevant points of view, including representatives of manufacturers of covered products, States, and efficiency advocates. 10 C.F.R. 6295(p)(4). DOE has interpreted "fairly representative of relative points of view" to mean that the group submitting a joint statement such as this one, where appropriate, include larger concerns and small businesses in the regulated industry/manufacturer community, energy advocates, energy utilities, consumers, and States. *See* 10 C.F.R. Part 430, Subpart C, Appendix A (Process Rule), section 10. The Process Rule states that it is necessary to determine the meaning of "fairly representative" on a case-by-case basis, subject to the circumstances of a particular rulemaking, to determine whether fewer or additional parties must be part of a joint statement in order to be "fairly representative of relevant points of view." *Id.*

The Joint Stakeholders are representative of a wide range of points of view as described below.

ACEEE, a nonprofit research organization, develops policies to reduce energy waste and combat climate change. Its independent analysis advances investments, programs, and behaviors that use energy more effectively and help build an equitable clean energy future.

ASAP organizes and leads a broad-based coalition effort that works to advance, win, and defend new appliance, equipment, and lighting standards that cut emissions that contribute to climate change and other environmental and public health harms, save water, and reduce economic and environmental burdens for low- and moderate-income households.

AHAM represents more than 150 member companies that manufacture 90% of the major, portable and floor care appliances shipped for sale in the U.S, including manufacturers of consumer room air cleaners. Home appliances are the heart of the home, and AHAM members provide safe, innovative, sustainable and efficient products that enhance consumers' lives. The home appliance industry is a significant segment of the economy, measured by the contributions of home appliance manufacturers, wholesalers, and retailers to the U.S. economy.

In all, the industry drives nearly $200 billion in economic output throughout the U.S. and manufactures products with a factory shipment value of more than $50 billion.

CFA is an association of more than 250 non-profit consumer and cooperative groups that was founded in 1968 to advance the consumer interest through research, advocacy, and education.

NRDC is an international nonprofit environmental organization with more than 3 million members and online activists. Since 1970, our lawyers, scientists, and other environmental specialists have worked to protect the world's natural resources, public health, and the environment. NRDC has offices in New York City, Washington, D.C., Los Angeles, San Francisco, Chicago, Bozeman, MT, and Beijing.

NYSERDA is a public benefit corporation and offers information and analysis, innovative programs, technical expertise, and support to help New Yorkers increase energy efficiency, save money, use renewable energy, and reduce reliance on fossil fuels. NYSERDA's mission is to advance clean energy innovation and investments to combat climate change, improving the health, resiliency, and prosperity of New Yorkers and delivering benefits equitably to all. NYSERDA works to help implement New York State's nation-leading climate agenda, which is the most aggressive climate and clean energy initiative in the nation; New York is advancing an orderly and just transition to clean energy that creates jobs and continues fostering a green economy as our communities continue recovering from the COVID-19 pandemic.

PG&E represents one of the largest combined gas and electric utilities in the Western U.S., serving over 16 million customers across northern and central California. As an energy company, we advocate for appliance efficiency standards to cut costs and reduce consumption while maintaining or increasing consumer utility of products. We have a responsibility to our customers to advocate for standards that accurately reflect the climate and conditions of our respective service areas.

III. <u>Rationale For The Negotiations.</u>

The Joint Stakeholders entered into discussions on a federal test procedure and energy conservation standards for consumer room air cleaners for three main reasons. First, it was thought that a negotiation might facilitate a joint recommendation on coverage and an applicable test procedure to streamline DOE's rulemaking process and on first-time energy conservation standards that DOE could efficiently adopt through a direct final rule. Second, these discussions allowed stakeholders to develop creative approaches, both regulatory and non-regulatory, which are more difficult to discuss and develop in a normal notice and comment rulemaking.[2] Third, a consistent national standard maximizes energy savings and minimizes costs to manufacturers and consumers. The Joint Stakeholders believe that all of these goals were achieved and will be borne out in the promulgation of the rules and their implementation in the future.

DOE has previously encouraged stakeholders to consider informal discussions, which could result in a consensus agreement. Furthermore, in 2007, Congress amended EPCA to expedite the

[2] Note that the Joint Stakeholders are also pursuing legislative implementation of our agreement.

rulemaking process by authorizing DOE to issue direct final rules establishing new energy conservation standards upon receipt of joint stakeholders' proposals.

The Joint Stakeholders entered into such discussions based on 42 U.S.C. § 6295(p)(4) and applicable sections of the Process Rule in 10 C.F.R. Part 430, Subpart C, Appendix A. The Process Rule states that DOE supports efforts by groups of interested parties to develop and present consensus recommendations on proposals for new or revised standards. According to the rule, the "Department seeks to encourage development of consensus proposals . . . for new or revised standards because standards with such broad-based support are likely to balance effectively the various interests affected by such standards." Process Rule, section 1(g). Further, the Process Rule indicates that, upon receipt of a joint proposal—including a consensus recommendation—submitted by "interested persons that are fairly representative of relevant points of view, DOE may issue a direct final rule (DFR) establishing energy conservation standards for a covered product . . . if DOE determines the recommended standard is in accordance with 42 U.S.C. 6295(o) . . ." Process Rule, section 10.

The Joint Stakeholders' proposal satisfies the criteria of the law and the Process Rule. It represents a consensus on standards, which are currently the maximum level that is technologically feasible and economically justified and which, of course, ultimately requires the exercise of judgment and discretion. <u>The Joint Stakeholders urge DOE to quickly finalize the recommended applicable test procedure and, concurrently, publish a direct final rule on the recommended standards by December 31, 2022 consistent with the recommended energy conservation standards. As discussed in Section VI, below, DOE should adopt the Joint Proposal for consumer room air cleaners</u>.

IV. The Negotiations Process

The parties' discussions commenced in 2021 and an agreement was finalized on July 22, 2022. Discussions were held, and empirically- and technically-based and justified proposals were made considering publicly available data from the ENERGY STAR program, the AHAM air cleaner directory,[3] consumer research studies on air cleaner ownership and distribution through NPD sales data by Evergreen Economics,[4] and data AHAM collected from its members. The Joint Stakeholders' proposal is economically justified, applying the relevant criteria in EPCA. *See* Section VI of these comments. These comments, however, specifically relate only to the consumer room air cleaner test procedure and energy conservation standards rulemaking and are not intended to create substantive "precedent" for other DOE appliance standards actions.

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[3] AHAM Verifide Directory, *available at* https://ahamverifide.org/directory-of-air-cleaners/.

[4] Evergreen Economics. (2022). Air Purifier NPD Analysis, Final Draft. Conducted for Energy Solutions and Pacific Gas and Electric Company (PG&E) Evergreen Economics. (2022). Air Purifier Study Results. Conducted for Pacific Gas and Electric Company (PG&E)

V. The Joint Stakeholder Proposal

A. Scope Of Coverage And Definitions

The Joint Proposal covers consumer room air cleaners, which may be less-inclusive than the broad air cleaner definition DOE adopted in 10 C.F.R. 430.2.[5] The Joint Stakeholders recommend DOE define *consumer room* air cleaner as follows:

> The term "consumer room air cleaner" means a consumer product for improving air quality that—
> a) is an electrically-powered, self-contained, mechanically encased assembly;
> b) contains means to remove, destroy, and/or deactivate particulates, volatile organic compounds (VOCs) and/or microorganisms, from the air;
> c) includes conventional room air cleaners;
> d) excludes central air conditioners, room air conditioners, portable air conditioners, dehumidifiers, and furnaces, as each is defined in in section 430.2 of title 10, Code of Federal Regulations (as currently in effect); and
> e) excludes duct type devices.

This definition is nearly identical to the definition DOE finalized for "air cleaner" in 10 C.F.R. 430.2. The main difference is that the above definition is specific to consumer *room* air cleaners only and, thus, excludes duct type devices. We note that the test procedure in the Joint Proposal only applies to the products identified in our proposed definition—i.e., it does not apply to duct type devices, whole-home devices, or those without a fan for air circulation. Nothing in the Joint Proposal relates to DOE's authority to regulate air cleaners as defined in 10 C.F.R. 430.2 other than those we have identified in this agreement as consumer room air cleaners.

As referenced in the definition for consumer room air cleaner, the Joint Stakeholders recommend DOE define the terms conventional room air cleaner and duct type devices as follows:

> The term "conventional room air cleaner" means a consumer room air cleaner that—
> a) is a portable or wall mounted (fixed) unit that plugs into an electrical outlet;
> b) operates with a fan for air circulation; and
> c) contains a means to remove, destroy, and/or deactivate particulates.
>
> The term "duct type devices" means an air cleaner designed and marketed for use in and at adjoining ducts, including the plenum, of heating, air conditioning, and ventilating systems. The air cleaner may not be provided with a fan, and the duct may provide part or all of the cabinet. This type of air cleaner may be either cord-and-plug connected or permanently connected to the electrical supply source.

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[5] Energy Conservation Program: Final Determination of Air Cleaners as a Covered Consumer Product; Docket No. EERE-20210BT-DET-0022; RIN 1904-AF25 (July 15, 2022).

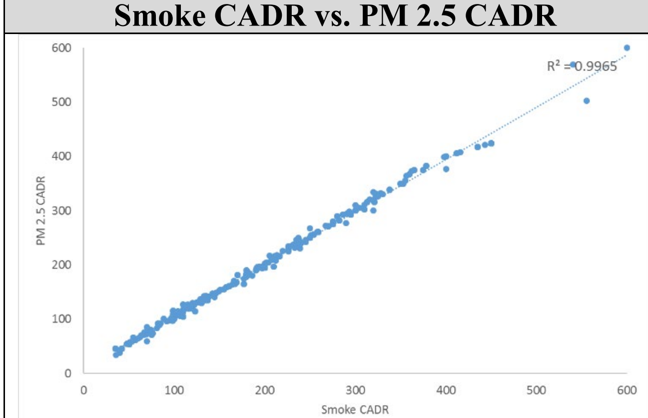
B. Test Procedure

For compliance with the standards in tier one and tier two of the Joint Proposal, the Joint Stakeholders recommend that DOE adopt in full AHAM AC-7-2022 (which is currently in final draft form, but we hope to soon finalize) as the test procedure. If a final version of AHAM AC-7-2022 is not available for incorporation by reference quickly enough, we still urge DOE to align with the final draft we are providing to DOE and AHAM authorizes DOE to use the text of the final draft as the basis for the Federal test procedure.

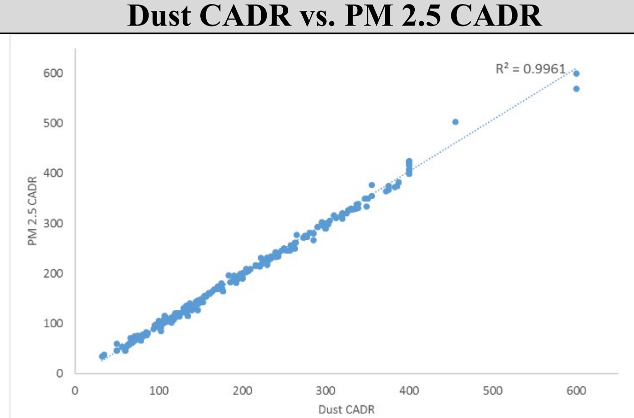
In addition, in order to ensure that, given the expedited compliance date of one year after a date of publication in the Federal Register for the recommended standards in tier one, assuming DOE publishes the final rule on December 31, 2022, the Joint Stakeholders also recommend that DOE permit section 6.2 of ANSI/AHAM AC-1-2020 for *Dust* Clean Air Delivery Rate (CADR) to be applied as an alternative for the purposes of calculating $PM_{2.5}$ CADR in AHAM AC-7-2022 for the tier one standards. Many products have already been tested per ANSI/AHAM AC-1-2020. The Dust CADR in that standard is nearly identical to the subset Dust CADR used to calculate the $PM_{2.5}$ CADR, as shown below. Allowing this alternative will ensure that manufacturers are not required to re-test using AHAM AC-7-2022 to demonstrate compliance with a new standard on such a short timeline.

PM 2.5 specifically refers to a category of fine particulate matter that is 2.5 microns or smaller in size. PM 2.5 CADR is the geometric mean of the measured cigarette smoke and dust CADR of an air cleaner and represents the rate at which a room air cleaner removes fine particulate matter from the air. Due to the small size, PM 2.5 particles can adequately represent a unit's performance for other larger particles as well. AHAM AC-7 in its current final draft form measures efficiency based on $PM_{2.5}$ CADR as the numerator. AHAM calculations of available data demonstrate a strong relationship between the $PM_{2.5}$ CADR calculation and measured Smoke CADR (R-squared of 99.6%) and measured Dust CADR(R-squared of 99.6%).

Smoke CADR vs. PM 2.5 CADR



Dust CADR vs. PM 2.5 CADR



Without the ability to rely on existing test data under AHAM AC-1-2020 for the proposed tier one standards, manufacturers will incur unnecessary cost to demonstrate compliance and will not be able to meet the incredibly quick recommended compliance date of December 31, 2023 which is integral to the Joint Proposal. This is because re-testing will take too long to allow manufacturers to certify products before they are imported into the United States, particularly given the well-documented supply chain challenges facing all consumer products.[6] Given that there is virtually no difference in results, there is no reason to require such costly and burdensome re-testing.

The Process Rule states that "DOE will adopt industry test procedure standards as DOE test procedures for covered products and equipment, but only if DOE determines that such procedures would not be unduly burdensome to conduct and would produce test results that reflect the energy efficiency . . . or estimated operating costs of that equipment during a representative average use cycle." Process Rule, section 8(c). This means that, so long as DOE is satisfied that the industry test procedure meets EPCA's requirements in 42 U.S.C. § 6293(b)(3), DOE will adopt it.

The Joint Stakeholders are united in their view that AHAM AC-7-2022 satisfies EPCA's criteria. Our agreement on energy conservation standards levels is premised on the test procedure as outlined in AHAM AC-7-2022 (in its draft and soon to be finalized form) and modification of that test may impact the Joint Stakeholders' continued support for the standards levels in the Joint Proposal.

The test procedure in the Joint Proposal is reasonably designed to produce test results that measure energy efficiency of consumer room air cleaners during a representative average use cycle. AHAM AC-7-2022 measures $PM_{2.5}$ CADR/Watt. Fine particulate matter has been shown to cause serious health problems and can get into the lungs and bloodstream. The Environmental Protection Agency (EPA), among other experts, has indicated that $PM_{2.5}$ particles are the primary concern from a health standpoint and are often found indoors.[7]

Moreover, the recommended metric incorporates standby energy using EPA ENERGY STAR's annual energy determination from the ENERGY STAR Version 2.0 room air cleaner specification. EPA's annual energy method which assumes 16 hours in active mode and eight hours in standby mode per day: Annual Energy (kWh) = ((Operating Watts x 5,840 hours) + (Standby watts x 2,920 hours)) / 1000.

The test procedure in the Joint Proposal is not unduly burdensome to conduct, so long as DOE permits the equivalent values as described above for tier one. The test procedure requires

[6] Supply chain challenges are making shipping time unpredictable and are causing significant delays in component parts and finished goods reaching consumers. Thus, it will be very challenging for manufacturers to ensure all products imported by the recommended compliance date comply with the recommended standards. Certainty in the regulation and flexibility in the testing are necessary to allow manufacturers to ensure compliance.

[7] *See* EPA, Residential Air Cleaners, A Technical Summary, 3d Ed. (July 2018), *available at* https://www.epa.gov/sites/default/files/2018-07/documents/residential_air_cleaners_-_a_technical_summary_3rd_edition.pdf; EPA; EPA, Guide to Air Cleaners in the Home, 2d Ed. (July 2018), *available at* https://www.epa.gov/sites/default/files/2018-07/documents/guide_to_air_cleaners_in_the_home_2nd_edition.pdf.

measurement only of dust and smoke CADR to calculate $PM_{2.5}$. The test procedure, on average, takes about three hours to complete and can be done with automated collection of the data.

We note that when DOE considers an amended test procedure per 42 U.S.C. § 6293(b)(1) and amended energy conservation standards per 42 U.S.C. § 6295(m)(1), the Joint Stakeholders recommend that DOE consider amending the test procedure to include measurement of automatic mode if such inclusion improves the accuracy and representativeness of energy use measurements.[8] AHAM's task force is currently working to establish such a test procedure and ASAP, DOE, and Guidehouse, are participating on that task force.

The Process Rule provides that new test procedures will be finalized at least 180 days prior to the close of the comment period for a proposed rule on energy conservation standards. *See* Process Rule, section 8(d)(1). It also provides for certain exceptions to that rule including that parties may submit a consensus recommendation and may specify a time period between finalization of the test procedure and the close of the comment period for a NOPR proposing new or amended energy conservation standards. *See* Process Rule, section 8(d)(2)(ii).

The policy undergirding the process requirement that test procedures be finalized well in advance of the end of a comment period on proposed new or amended energy conservation standards is that manufacturers and other interested parties require time to test products in order to fully understand the impact of proposed energy conservation standards. And the exception that the typical 180-day period between the finalization of a test procedure and the end of a comment period on a proposed standard may be shortened if recommended by a consensus agreement recognizes that, in cases where the parties agree to a test procedure and/or energy conservation standards, those impacts are already well-understood.

In this case, the Joint Stakeholders have worked together on the development of an applicable test procedure. The procedure is based on previously existing test procedures for consumer room air cleaners, including the procedure used for the ENERGY STAR program. Thus, the procedure is well-understood by manufacturers and other stakeholders and additional time is not needed to test products in order to assess the impact of standards. In negotiating the agreement, the Joint Stakeholders have already been able to assess the potential impact of the agreed upon standards using the recommended test procedure. Thus, we urge DOE to rely upon the exception in section 8(d)(2)(ii) of the Process Rule to quickly finalize the test procedure. The Joint Stakeholders recommend that there need not be any time between the finalization of the test procedure and the end of the comment period on or even the issuance of a direct final rule on energy conservation standards for consumer room air cleaners provided that both the test procedure and standards are consistent with the recommendations in the Joint Proposal.

[8] 42 U.S.C. § 6293(b)(1) requires DOE to review test procedures once every seven years. 42 U.S.C. § 6295(m)(1) requires DOE, no later than six years after issuance of a final rule establishing or amending a standard, to issue a notice of proposed rulemaking amending standards or a notice of determination that amended standards do not need to be amended.

C. Energy Conservation Standards

The specific standards levels the Joint Stakeholders propose are found below. Significantly, the proposals will be the first mandatory federal test procedure and energy conservation standards for this product category and, as such, represent significantly lower energy use. The proposed standards in tier one are equivalent to the standards levels a limited number of states have adopted,[9] and the proposed standards in tier two are equivalent to the ENERGY STAR Version 2.0 specification and standards adopted by Washington State.[10] Importantly, central to this agreement are the proposed compliance dates—December 31, 2023 for tier one and December 31, 2025 for tier two. The significantly expedited compliance dates achieve significant national savings earlier than through the usual DOE rulemaking process and provide for consistent, national standards in place of state-by-state regulation. The proposed standards are estimated to save 1.9 quads of full-fuel-cycle energy nationally over 30 years of sales.[11]

Joint Stakeholder Agreement
Consumer Room Air Cleaners

Product Description	Tier 1 New Standard (IEF) Dec. 31, 2023*	Tier 2 Standard (IEF) Dec. 31, 2025*
$PM_{2.5}$CADR of 10 or greater and less than 100	1.69 $PM_{2.5}$ CADR/Watt	1.89 $PM_{2.5}$ CADR/Watt
$PM_{2.5}$CADR of 100 or greater and less than 150	1.90 $PM_{2.5}$ CADR/Watt	2.39 $PM_{2.5}$ CADR/Watt
$PM_{2.5}$CADR of 150 or greater	2.01 $PM_{2.5}$ CADR/Watt	2.91 $PM_{2.5}$ CADR/Watt
*Assuming DOE publishes final rule by Dec. 31, 2022. Otherwise, any of the Joint Stakeholders may withdraw support for the recommendations in the Joint Proposal.		

Notably, the first product class in the Joint Proposal begins at a $PM_{2.5}$ CADR of 10. This is lower (and more expansive) than the ENERGY STAR Version 2.0 threshold and is intended to capture tabletop/desk portable room air cleaners. Per AHAM's Certification Program, a suggested room

[9] The District of Columbia, Maryland, Nevada, and New Jersey have passed laws with standards as follows:

Smoke CADR Bins	Minimum Smoke CADR/W
$30 \leq CADR < 100$	1.7
$100 \leq CADR < 150$	1.9
$CADR \geq 150$	2.0

[10] The Joint Stakeholders converted the state standard levels and the ENERGY STAR/Washington State levels to $PM_{2.5}$ CADR/Watt. The conversion method is included at Appendix A. Notably, Washington State's standards are scheduled to be effective on January 1, 2024. The recommendations in this proposal will delay those standards by about two years for Washington to December 31, 2025.

[11] We will provide our calculations to Guidehouse.

size is the smoke CADR multiplied by 1.55. Examples: (1) 10 CADR x 1.55 = 15.5 sq.ft. room, (2) 30 CADR x 1.55 = 46.5 sq.ft. room.[12]

The Joint Proposal recommends the above product classes which are consistent with the product classes in the ENERGY STAR Version 2.0 specification. In developing that specification, EPA justified these product classes by reasoning that they are necessary to retain consumer utility. EPA stated that it "heard from stakeholders that smaller-CADR products, that offer a lower cost option for small rooms, have more difficulty achieving ENERGY STAR than larger-CADR products. EPA believes the proposal to set criteria based on CADR will ensure consumers have a wide range of choices in each size bin of room air cleaners and will not inadvertently choose a product that is under or oversized for their space."[13] Smaller CADR products, which are appropriate for small rooms, have fewer available design options for reaching higher levels of efficiency.

It is more difficult for smaller room air cleaners to reach higher levels of efficiency because physics is working in the opposite direction. Smaller products require smaller components, like smaller fan blades, to fit in the smaller "box." This makes achieving higher levels of efficiency a more difficult design challenge while retaining the utility of the smaller size. For example, to lower wattage in an effort to increase the CADR/Watt, a smaller room air cleaner could need a more efficient blade design or a more efficient motor.

As the blade design is made more efficient despite its smaller diameter, the optimization point is tight to achieve adequate air movement while not increasing noise levels beyond a tolerable level. With the smaller size, a design that delivers less air flow or higher sound level—and thus does not deliver consumer utility—may be the only options available.

Were smaller products required to meet the same efficiency levels as larger and higher CADR/Watt models, a greater change in efficiency of the motor would be necessary. That degree of improvement—e.g., ten percent—for a product already operating at, for example 50 watts or less, is quite challenging and could require more expensive motor technology that could lead to standards that are not economically justified. Alternatively, a manufacturer could slow down the motor to reduce wattage which would also reduce the air flow and the CADR values—thus, the CADR/Watt ratio would not change, or could even be less efficient.

Thus, from a technological perspective, the recommended product classes will help ensure that a broad range of capacity changes remain available for consumers.

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[12] *See* AHAM, Frequently Asked Questions about Testing of Portable Air Cleaners, at 5, *at* https://ahamverifide.org/wp-content/uploads/2020/09/Testing-of-Portable-Air-Cleaner-Performance-FAQs-2020-Updates.pdf.

[13] EPA, ENERGY STAR Product Specification for Room Air Cleaners, Eligibility Criteria, Draft 1 Version 2.0 (Mar. 18, 2019), *available at* https://www.energystar.gov/sites/default/files/asset/document/ENERGY%20STAR%20Room%20Air%20Cleaners%20Draft%201%20V2.0%20Specification.pdf.

The Joint Stakeholders are united in their view that the Joint Proposal comports fully with the standards-setting criteria in EPCA and the Joint Proposal was designed to achieve the maximum improvement in energy efficiency that is technologically feasible and economically justified as required by 42 U.S.C. § 6295(o). The Joint Proposal proposes standards that would decrease maximum energy use of a covered product in both tier one and tier two, and thus complies with EPCA's prohibition against standards that increase maximum allowable energy use of a covered product. 42 U.S.C. § 6295(o)(1).

The Joint Proposal, if promulgated as a standard, will result in benefits that exceed the burdens imposed to the greatest extent practicable taking into account the desirability of mitigating manufacturer burdens and allowing for marketplace innovation. 42 U.S.C. § 6295(o)(2)(B)(i).

Below, the Joint Stakeholders detail the ways in which the Joint Proposal meets EPCA's requirements of being the most stringent standard level which is technically feasible and economically justified, taking into account the societal, consumer, and manufacturer interests set forth in 42 U.S.C. § 6295(o).

i. *Technical Feasibility*

The Joint Stakeholders determined that the standards levels in the Joint Proposal can be reached through existing technology. There currently are products on the market that meet the proposed standards levels for both tier one and tier two in all proposed product classes.

ii. *Economic Impact On Consumers*

The Joint Stakeholders are confident that the Joint Proposal's economic effect on consumers is justified as required by 42 U.S.C. § 6295(o)(2)(B)(i)(I). It is not likely that product changes will require platform changes to meet the recommended standards levels. As those platform-level changes which require retooling are the most costly, we do not expect a significant number of consumers to experience a net cost nor do we expect low-income consumers or other subgroups to experience a disproportionate impact.

iii. *Economic Impact On Manufacturers*

The Joint Stakeholders agree that the Joint Proposal will have acceptable impacts on manufacturers. This conclusion is endorsed by all of the manufacturers signing this agreement, including both large and small manufacturers and "full line" and "niche" companies. That manufacturers can use a well-understood test procedure and not need to engage in re-testing to meet tier one standards will helps to mitigate the burden on manufacturers.

Although the impacts on manufacturers are "acceptable" at the proposed standard levels, at higher efficiency levels, additional economic impacts on manufacturers are likely to occur. Investments will be higher for higher levels of efficiency.

The Joint Stakeholders include more than sixty percent of manufacturers of products covered by the Joint Proposal. The Joint Stakeholders, therefore, believe that the collective statement of manufacturers that the economic impacts of the Joint Proposal are acceptable should address the Economic Impact on Manufacturers criteria, and that DOE need not conduct a more detailed manufacturer impact assessment. We believe that a more detailed analysis would be an unnecessary use of DOE and manufacturer resources and time. While we strongly urge that DOE not conduct its detailed manufacturer impact assessment, if DOE decides it needs to conduct such an assessment, the Joint Stakeholders are prepared to fully cooperate, including scheduling urgent manufacturer interviews.

iv. *Life Cycle Costs*

The Joint Stakeholders believe that consumers will recoup additional upfront costs through energy savings before the end of the product's useful life. The average electricity bill savings for affected consumers will be approximately $23 per year from the recommended tier one standards and an additional $21 from the recommended tier two standards, and we believe the simple payback to consumers will be less than the product's average useful life, which we are assuming to be nine years for the purposes of our analysis.[14]

v. *Energy Savings*

The Joint Proposal for consumer room air cleaners is estimated to save 1.9 quads of full-fuel-cycle energy over 30 years of sales based on our analysis. The energy savings provided by the Joint Proposal are significant for this product category.[15]

vi. *Lessening Of Utility Or Performance Or Availability of Products*

The Joint Proposal will not result in significant lessening of utility or performance or availability of the covered products as prohibited by EPACT under the so-called "safe harbor" exception. 42 U.S.C. § 6295(o)(2)(B)(IV). The Joint Proposal was specifically designed to maintain product design diversity and utility in the marketplace for consumer room air cleaners, and, therefore, deals with utility, performance, and availability-related concerns that could have resulted if the standard were set at a different level and/or with different boundaries between product classes.

vii. *Impact Of Lessening Of Competition*

The Joint Stakeholders do not believe the Joint Proposal would support a Department of Justice determination that the standard would lead to the likelihood of reduced competition. 42 U.S.C. § 6295(o)(2)(B)(V). The Joint Proposal was developed in consultation with manufacturers of consumer room air cleaners, including large, medium, and small manufacturers, and has been

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[14] *See* https://www.xcelenergy.com/staticfiles/xe/PDF/Regulatory/CO-Rates-and-Regs-DSM-Cadmus-RPP-Product-Analysis-August-2015.pdf.

[15] The Joint Stakeholders recognize that DOE has indicated that whether savings are "significant" is to be determined on a case-by-case basis and our joint statement that 1.9 quads is a significant savings of energy is specific to this particular rulemaking. Details of the Joint Stakeholders estimate is provided in a separate spreadsheet for reference.

designed to mitigate any negative competitive impacts. The Joint Proposal is not expected to eliminate any competitors.

viii. *Need Of The Nation To Conserve Energy*

As noted above, the energy savings from the Joint Proposal with regard to consumer room air cleaners are estimated to save approximately 1.9 quads of full-fuel-cycle energy over 30 years of sales. These savings will in turn result in reduced emissions of carbon dioxide.

VI. <u>Conclusion</u>

The Joint Stakeholders request that DOE propose and finalize a test procedure that adopts the test procedures we recommend above as soon as possible. And we request that DOE finalize energy conservation standards consistent with our above recommendations. Specifically, the Joint Stakeholders recommend that the DOE publish a direct final rule on standards by December 31, 2022, adopting the consumer room air cleaner standards and compliance dates contained in the Joint Proposal. A final rule later than that will make the compliance date we propose—December 31, 2023—very difficult for manufacturers to implement and, thus, will unravel the thread that ties our agreement together. Thus, should DOE not act by that date, some or all of the Joint Stakeholders may withdraw support for this agreement and Joint Proposal. We believe that the broad consensus in support of the proposed standards will allow DOE to move directly to a final rule, avoiding lost energy savings and preserving DOE resources for other rulemakings.

Joint Stakeholders

Kevin Messner
Sr. Vice President, Policy and Government Relations
Association of Home Appliance Manufacturers

Andrew deLaski
Executive Director
Appliance Standards Awareness Project

On Behalf of—

Members of AHAM's Portable Appliance Division That Manufacture Consumer Room Air Cleaners:

3M Company

Access Business Group, LLC
ACCO Brands Corporation

Air King, Air King Ventilation Products

Airgle Corporation
Beijing Smartmi Electronic Technology Co., Ltd.

American Council for an Energy-Efficient Economy
Consumer Federation of America
Natural Resources Defense Council
New York State Energy Research Development Authority
Pacific Gas and Electric Company

BISSELL Inc.
Blueair Inc.
De'Longhi America, Inc.
Dyson Limited
Fellowes Inc.
Field Controls
GE Appliances, a Haier company
Gree Electric Appliances, Inc.
Groupe SEB USA
Guardian Technologies, LLC
Helen of Troy - Health & Home
iRobot
Lasko Products, Inc.
Molekule Inc.
Newell Brands, Inc.
Oransi
SharkNinja Operating, LLC
Sharp Electronics Corporation
Sharp Electronics of Canada Ltd.
Sunbeam Products, Inc.
Trovac Industries Ltd
Vornado Air, LLC
Whirlpool Corporation
Winix Inc.
Zojirushi America Corp.

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Appendix A

Joint Stakeholders Smoke CADR / W to IEF (PM 2.5 CADR / W) Conversion Methodology

In order to convert current state regulations and ENERGY STAR specification levels from Smoke CADR / W to IEF (PM 2.5 CADR / W), the following sources were used:

1. Current Regulations:

Product Classes	State Regulations Smoke CADR/W	ENERGY STAR Smoke CADR/W
30 ≤ CADR < 100	≥1.7	≥1.9
100 ≤ CADR < 150	≥1.9	≥2.4
CADR ≥ 150	≥2.0	≥2.94

*Note: ENERGY STAR specification uses the same CADR segments

2. Annual energy was determined for each segment using EPA's annual energy calculation and the lower and upper CADR boundaries:

   Annual Energy = ((Smoke CADR (boundary)/ Smoke CADR/W*5840) + (Standby Power*2920 hours))/1000

   Standby Power = 1.4W (average used based on manufacturer data)
   5840 = 16 active hours
   2920 = 8 standby hours
   / 1000 = converting Wh to kWh

3. Conversion of Smoke CADR to PM 2.5 CADR using average (AHAM Verifide (model) Directory) = **1.01027**

4. Average smoke CADR by bin used to derive annual energy to determine the Integrated Energy Factor (IEF) are from consumer research study and AHAM data

Product Classes	Smoke CADR
30 ≤ CADR < 100	87
100 ≤ CADR < 150	113
CADR ≥ 150	253

5. IEF is determined using the below calculation, from AHAM AC-7-2022:

$$IEF = \left[\frac{CADR\left(\frac{ft^3}{min}\right)}{\left(AEC\left(\frac{kWh}{year}\right) * \frac{1\ year}{5840\ hours} * \frac{1000\ Wh}{1\ kWh}\right)}\right]$$

$$IEF\ (PM_{2.5}\ CADR/watt) = \left[\frac{CADR}{\left(\frac{AEC}{5.840}\right)}\right]$$

6. Final conversion from Smoke CADR/W levels to IEF (PM 2.5 CADR/W):

Recommended Tier	**Product Classes**	**Average Smoke CADR**	**Smoke CADR/W Levels (District of Columbia, Maryland, Nevada, New Jersey Levels)**	**Per-Unit Annual Energy**	**IEF PM 2.5 CADR/W**
Tier 1	$30 \leq CADR < 100$	87	1.7	303	1.69
	$100 \leq CADR < 150$	113	1.9	351	1.90
	$CADR \geq 150$	253	2.0	743	2.01

Recommended Tier	**Product Classes**	**Average Smoke CADR**	**Smoke CADR/W Levels (V. 2.0 ENERGY STAR and Washington State Levels)**	**Per-Unit Annual Energy**	**IEF PM 2.5 CADR/W**
Tier 2	$30 \leq CADR < 100$	87	1.9	271	1.89
	$100 \leq CADR < 150$	113	2.4	279	2.39
	$CADR \geq 150$	253	2.9	514	2.91